

AP[®] Calculus AB Practice Exam

From the 2015 Administration

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A A

SECTION I, Part A

Time—55 minutes

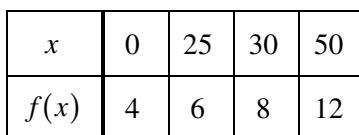
Number of questions—28

A CALCULATOR MAY NOT BE USED ON THIS PART OF THE EXAM.

Directions: Solve each of the following problems, using the available space for scratch work. After examining the form of the choices, decide which is the best of the choices given and fill in the corresponding circle on the answer sheet. No credit will be given for anything written in the exam book. Do not spend too much time on any one problem.

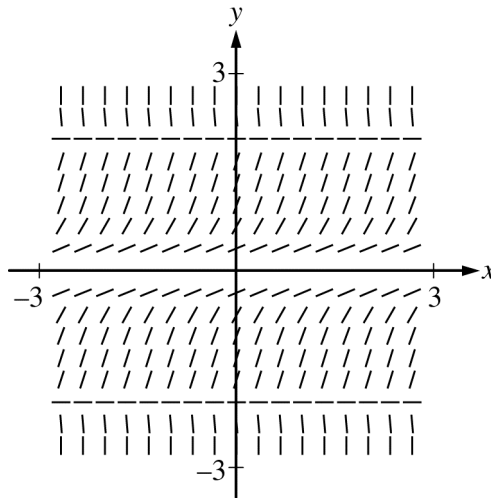
In this exam:

- (1) Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.
- (2) The inverse of a trigonometric function f may be indicated using the inverse function notation f^{-1} or with the prefix “arc” (e.g., $\sin^{-1} x = \arcsin x$).



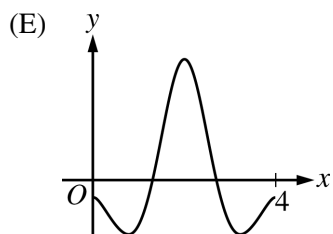
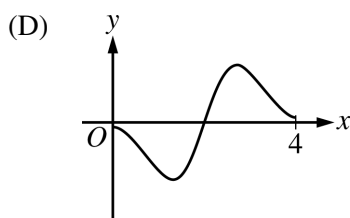
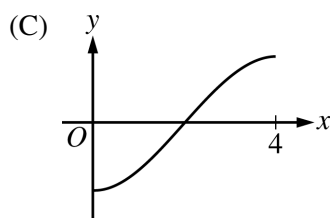
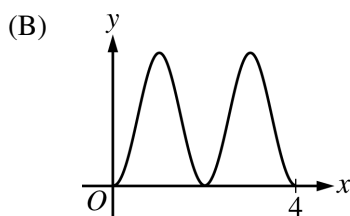
- $$f(x) = \begin{cases} x^2 \sin(\pi x) & \text{for } x < 2 \\ x^2 + cx - 18 & \text{for } x \geq 2 \end{cases}$$

- 6-**



9. If $f''(x) = x(x+2)^2$, then the graph of f is concave up for
- (A) $x < 0$
(B) $x > 0$
(C) $-2 < x < 0$
(D) $x < -2$ and $x > 0$
(E) all real numbers

- (A) 



20. Let f be the function given by $f(x) = \frac{(x-4)(2x-3)}{(x-1)^2}$. If the line $y = b$ is a horizontal asymptote to the graph

(A) 0 (B) 1 (C) 2 (D) 3 (E) 4

(A) $\frac{2}{3}$ (B) $\frac{4}{3}$ (C) 2 (D) $\frac{2\pi}{3}$ (E) $\frac{4\pi}{3}$

22. Let f be the function given by $f(x) = \frac{kx}{x^2 + 1}$, where k is a constant. For what values of k , if any, is f strictly decreasing on the interval $(-1, 1)$?

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any part of this page is illegal.**

24. Which of the following is the solution to the differential equation $\frac{dy}{dx} = 5y^2$ with the initial condition

(A) $y = \sqrt{9e^{5x}}$

(B) $y = \sqrt{\frac{1}{9}e^{5x}}$

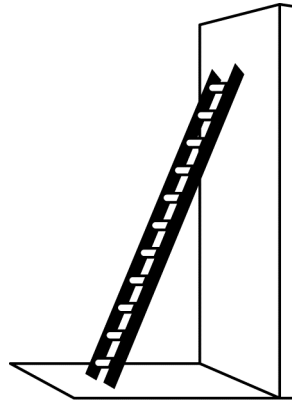
(C) $y = \sqrt{e^{5x} + 9}$

(D) $y = \frac{3}{1 - 15x}$

(E) $y = \frac{3}{1 + 15x}$

25. $\lim_{h \rightarrow 0} \frac{\sin\left(\frac{\pi}{3} + h\right) - \sin\left(\frac{\pi}{3}\right)}{h}$ is

- (A) 0 (B) $\frac{\pi}{6}$ (C) $\frac{2}{3}$ (D) $\frac{\pi}{3}$ (E) $\frac{4}{3}$



- (A) $-\frac{2}{9}$ (B) $-\frac{1}{6}$ (C) $-\frac{2}{25}$ (D) $\frac{2}{25}$ (E) $\frac{1}{9}$

-21-

B**B****B****B****B****B****B****B****B****CALCULUS AB****SECTION I, Part B****Time—50 minutes****Number of questions—17**

A GRAPHING CALCULATOR IS REQUIRED FOR SOME QUESTIONS ON
THIS PART OF THE EXAM.

Directions: Solve each of the following problems, using the available space for scratch work. After examining the form of the choices, decide which is the best of the choices given and fill in the corresponding circle on the answer sheet. No credit will be given for anything written in the exam book. Do not spend too much time on any one problem.

BE SURE YOU ARE USING PAGE 3 OF THE ANSWER SHEET TO RECORD YOUR ANSWERS TO QUESTIONS NUMBERED 76–92.

YOU MAY NOT RETURN TO PAGE 2 OF THE ANSWER SHEET.

In this exam:

- (1) The exact numerical value of the correct answer does not always appear among the choices given. When this happens, select from among the choices the number that best approximates the exact numerical value.
- (2) Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.
- (3) The inverse of a trigonometric function f may be indicated using the inverse function notation f^{-1} or with the prefix “arc” (e.g., $\sin^{-1}x = \arcsin x$).

B**B****B****B****B****B****B****B****B**

76. The function $P(t)$ models the population of the world, in billions of people, where t is the number of years since January 1, 2010. Which of the following is the best interpretation of the statement $P'(1) = 0.076$?
- (A) On February 1, 2010, the population of the world was increasing at a rate of 0.076 billion people per year.
 - (B) On January 1, 2011, the population of the world was increasing at a rate of 0.076 billion people per year.
 - (C) On January 1, 2011, the population of the world was 0.076 billion people.
 - (D) From January 1, 2010 to January 1, 2011, the population of the world was increasing at an average rate of 0.076 billion people per year.
 - (E) When the population of the world was 1 billion people, the population of the world was increasing at a rate of 0.076 billion people per year.

B**B****B****B****B****B****B****B****B**

x	0	2	4	6	8	10
$f(x)$	5	7	8	0	-15	-20

77. Let f be a differentiable function with selected values given in the table above. What is the average rate of change of f over the closed interval $0 \leq x \leq 10$?

- (A) -6 (B) $-\frac{5}{2}$ (C) -2 (D) $-\frac{2}{5}$ (E) $\frac{2}{5}$

78. The rate at which motor oil is leaking from an automobile is modeled by the function L defined by $L(t) = 1 + \sin(t^2)$ for time $t \geq 0$. $L(t)$ is measured in liters per hour, and t is measured in hours. How much oil leaks out of the automobile during the first half hour?

- (A) 1.998 liters
(B) 1.247 liters
(C) 0.969 liters
(D) 0.541 liters
(E) 0.531 liters

B**B****B****B****B****B****B****B****B**

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
0	3	4	2	π

79. The table above gives values of the differentiable functions f and g and their derivatives at $x = 0$.

If $h(x) = \frac{f(x)}{g(x)}$, what is the value of $h'(0)$?

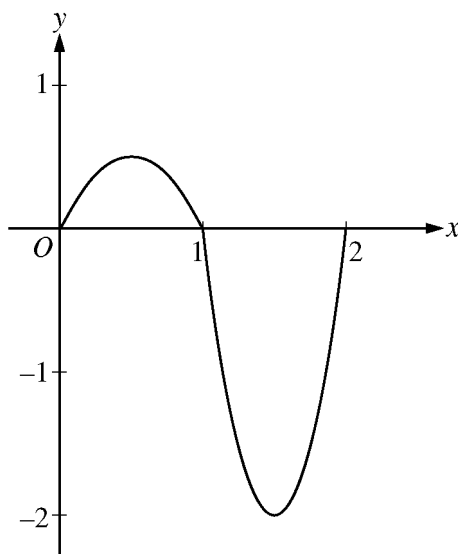
(A) $\frac{8 - 3\pi}{4}$

(B) $\frac{3\pi - 8}{4}$

(C) $\frac{4}{\pi}$

(D) $\frac{2 - 3\pi}{2}$

(E) $\frac{8 + 3\pi}{4}$

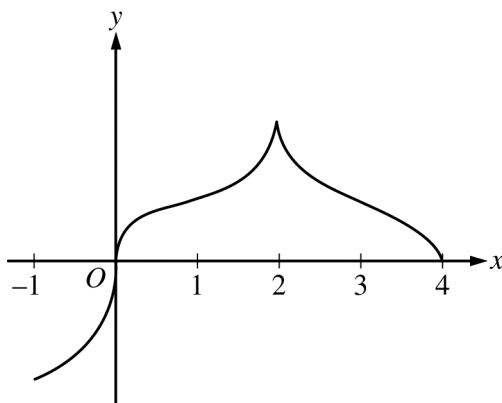
B**B****B****B****B****B****B****B****B**Graph of f'

80. The figure above shows the graph of f' , the derivative of a function f , for $0 \leq x \leq 2$. What is the value of x at which the absolute minimum of f occurs?

- (A) 0 (B) $\frac{1}{2}$ (C) 1 (D) $\frac{3}{2}$ (E) 2

81. What is the area of the region enclosed by the graphs of $y = \sqrt{4x - x^2}$ and $y = \frac{x}{2}$?

- (A) 1.707 (B) 2.829 (C) 5.389 (D) 8.886 (E) 21.447

B**B****B****B****B****B****B****B****B**Graph of f'

82. The graph of f' , the derivative of f , is shown above. The line tangent to the graph of f' at $x = 0$ is vertical, and f' is not differentiable at $x = 2$. Which of the following statements is true?
- (A) f' does not exist at $x = 2$.
- (B) f is decreasing on the interval $(2, 4)$.
- (C) The graph of f has a point of inflection at $x = 2$.
- (D) The graph of f has a point of inflection at $x = 0$.
- (E) f has a local maximum at $x = 0$.

B**B****B****B****B****B****B****B****B**

83. A particle moves along the x -axis so that its position at time $t > 0$ is given by $x(t)$ and

$$\frac{dx}{dt} = -10t^4 + 9t^2 + 8t. \text{ The acceleration of the particle is zero when } t =$$

- (A) 0.387 (B) 0.831 (C) 1.243 (D) 1.647 (E) 8.094

84. The function f is continuous on the closed interval $[1, 7]$. If $\int_1^7 f(x) \, dx = 42$ and $\int_7^3 f(x) \, dx = -32$,

then $\int_1^3 2f(x) \, dx =$

- (A) -148 (B) 10 (C) 12 (D) 20 (E) 148

B**B****B****B****B****B****B****B****B**

85. Let $y = f(x)$ define a twice-differentiable function and let $y = t(x)$ be the line tangent to the graph of f at $x = 2$. If $t(x) \geq f(x)$ for all real x , which of the following must be true?

(A) $f(2) \geq 0$

(B) $f'(2) \geq 0$

(C) $f'(2) \leq 0$

(D) $f''(2) \geq 0$

(E) $f''(2) \leq 0$

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86. The vertical line $x = 2$ is an asymptote for the graph of the function f . Which of the following statements must be false?

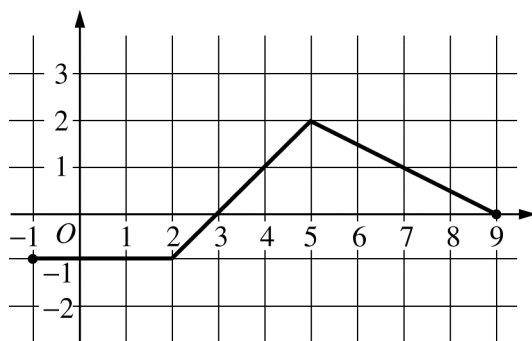
(A) $\lim_{x \rightarrow 2} f(x) = 0$

(B) $\lim_{x \rightarrow 2} f(x) = -\infty$

(C) $\lim_{x \rightarrow 2} f(x) = \infty$

(D) $\lim_{x \rightarrow \infty} f(x) = 2$

(E) $\lim_{x \rightarrow \infty} f(x) = \infty$

B**B****B****B****B****B****B****B****B**Graph of f

87. The graph of the piecewise linear function f is shown above. Let h be the function given by $h(x) = \int_{-1}^x f(t) dt$.

On which of the following intervals is h increasing?

- (A) $[-1, 3]$
- (B) $[0, 5]$
- (C) $[2, 5]$ only
- (D) $[2, 9]$
- (E) $[3, 9]$ only

B**B****B****B****B****B****B****B****B**

88. The first derivative of the function f is given by $f'(x) = \sin(x^2)$. At which of the following values of x does f have a local minimum?

- (A) 2.507 (B) 2.171 (C) 1.772 (D) 1.253 (E) 0

89. If $\lim_{x \rightarrow a} f(x) = f(a)$, then which of the following statements about f must be true?

- (A) f is continuous at $x = a$.
(B) f is differentiable at $x = a$.
(C) For all values of x , $f(x) = f(a)$.
(D) The line $y = f(a)$ is tangent to the graph of f at $x = a$.
(E) The line $x = a$ is a vertical asymptote of the graph of f .

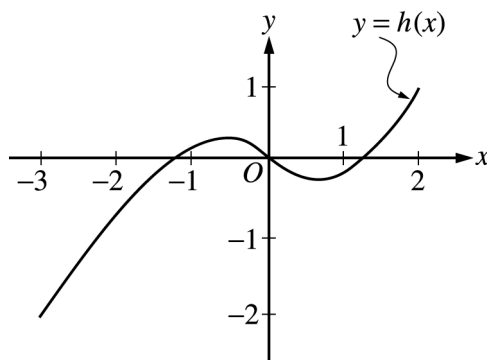
B**B****B****B****B****B****B****B****B**

90. The temperature F , in degrees Fahrenheit ($^{\circ}\text{F}$), of a cup of coffee t minutes after it is poured is given by $F(t) = 72 + 118e^{-0.093t}$. To the nearest degree, what is the average temperature of the coffee between $t = 0$ and $t = 10$ minutes?

- (A) 93°F
(B) 119°F
(C) 146°F
(D) 149°F
(E) 154°F
-

91. If $f'(x) = \cos(x^2)$ and $f(3) = 7$, then $f(2) =$

- (A) 0.241 (B) 5.831 (C) 6.416 (D) 6.759 (E) 7.241

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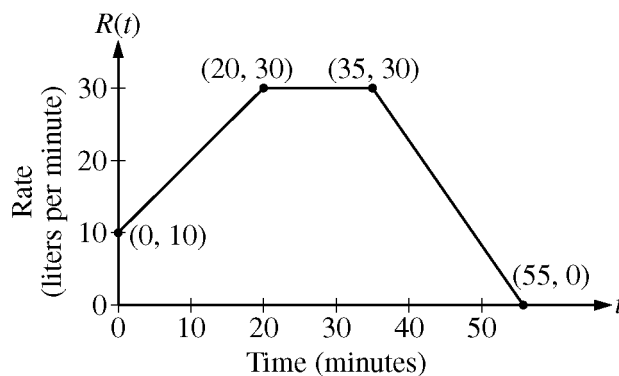
92. The graph of the function h is shown in the figure above. Of the following, which has the greatest value?

- (A) Average value of h over $[-3, 2]$
- (B) Average rate of change of h over $[-3, 2]$
- (C) $\int_{-3}^2 h(x) dx$
- (D) $\int_{-3}^0 h(x) dx$
- (E) $h'(0)$

CALCULUS AB
SECTION II, Part A
Time—30 minutes
Number of problems—2

A graphing calculator is required for these problems.

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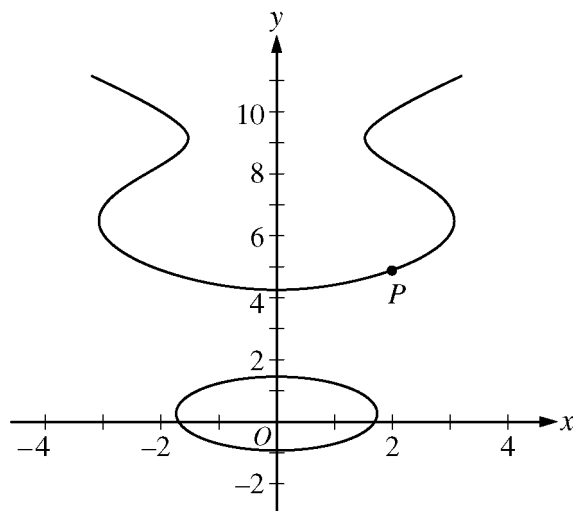
1. At time $t = 0$ minutes, a tank contains 100 liters of water. The piecewise-linear graph above shows the rate $R(t)$, in liters per minute, at which water is pumped into the tank during a 55-minute period.
- (a) Find $R'(45)$. Using appropriate units, explain the meaning of your answer in the context of this problem.

- (b) How many liters of water have been pumped into the tank from time $t = 0$ to time $t = 55$ minutes? Show the work that leads to your answer.

1**1****1****1****1****1****1****1****1****1**

- (c) At time $t = 10$ minutes, water begins draining from the tank at a rate modeled by the function D , where $D(t) = 10e^{(\sin t)/10}$ liters per minute. Water continues to drain at this rate until time $t = 55$ minutes. How many liters of water are in the tank at time $t = 55$ minutes?

-
- (d) Using the functions R and D , determine whether the amount of water in the tank is increasing or decreasing at time $t = 45$ minutes. Justify your answer.



2. The graph of the equation $x^2 = -2 + y + 5 \cos y$ is shown above for $y \leq 11$. It is known that $\frac{dy}{dx} = \frac{2x}{1 - 5 \sin y}$. The x -coordinate of point P shown on the graph is 2.
- (a) Write an equation for the line tangent to the graph at point P .

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- (b) For $y \leq 11$, find the y -coordinate of each point on the graph where the line tangent to the graph at that point is vertical.

-
- (c) Find the average value of the x -coordinates of the points on the graph in the first quadrant between $y = 5$ and $y = 9$.

CALCULUS AB
SECTION II, Part B
Time—60 minutes
Number of problems—4

No calculator is allowed for these problems.

DO NOT BREAK THE SEALS UNTIL YOU ARE TOLD TO DO SO.

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NO CALCULATOR ALLOWED

t (seconds)	0	3	5	8	12
$k(t)$ (feet per second)	0	5	10	20	24

3. Kathleen skates on a straight track. She starts from rest at the starting line at time $t = 0$. For $0 < t \leq 12$ seconds, Kathleen's velocity k , measured in feet per second, is differentiable and increasing. Values of $k(t)$ at various times t are given in the table above.
- (a) Use the data in the table to estimate Kathleen's acceleration at time $t = 4$ seconds. Show the computations that lead to your answer. Indicate units of measure.

-
- (b) Use a right Riemann sum with the four subintervals indicated by the data in the table to approximate $\int_0^{12} k(t) dt$. Indicate units of measure. Is this approximation an overestimate or an underestimate for the value of $\int_0^{12} k(t) dt$? Explain your reasoning.

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NO CALCULATOR ALLOWED

- (c) Nathan skates on the same track, starting 5 feet ahead of Kathleen at time $t = 0$. Nathan's velocity, in feet per second, is given by $n(t) = \frac{150}{t+3} - 50e^{-t}$. Write, but do not evaluate, an expression involving an integral that gives Nathan's distance from the starting line at time $t = 12$ seconds.

-
- (d) Write an expression for Nathan's acceleration in terms of t .

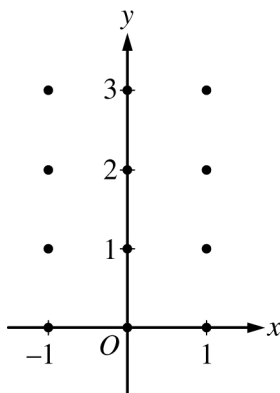
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NO CALCULATOR ALLOWED

4. Consider the differential equation $\frac{dy}{dx} = \frac{x(y-1)}{4}$.

(a) On the axes provided, sketch a slope field for the given differential equation at the twelve points indicated.



(b) Let $y = f(x)$ be the particular solution to the differential equation with the initial condition $f(1) = 3$. Write an equation for the line tangent to the graph of f at the point $(1, 3)$ and use it to approximate $f(1.4)$.

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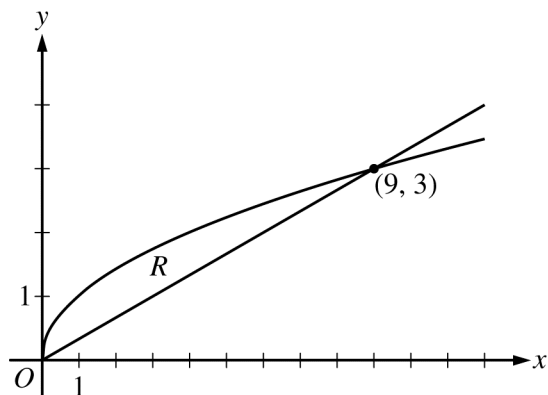
NO CALCULATOR ALLOWED

- (c) Find the particular solution $y = f(x)$ to the given differential equation with the initial condition $f(1) = 3$.

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NO CALCULATOR ALLOWED



5. Let R be the region in the first quadrant enclosed by the graphs of $g(x) = \sqrt{x}$ and $h(x) = \frac{x}{3}$, as shown in the figure above.
- (a) Find the area of region R .

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NO CALCULATOR ALLOWED

- (b) Write, but do not evaluate, an expression involving one or more integrals that gives the volume of the solid generated when R is revolved about the horizontal line $y = 4$.

-
- (c) Find the maximum vertical distance between the graph of g and the graph of h between $x = 0$ and $x = 16$. Justify your answer.

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6**6****6****6****6****6****6****6****6****6****NO CALCULATOR ALLOWED**

6. Let $g(x) = 4(x+1)^{-2/3}$ and let f be the function defined by $f(x) = \int_0^x g(t) dt$ for $x \geq 0$.

(a) Find $f(26)$.

(b) Determine the concavity of the graph of $y = f(x)$ for $x > 0$. Justify your answer.

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6**6****6****6****6****6****6****6****6****6****NO CALCULATOR ALLOWED**

- (c) Let h be the function defined by $h(x) = x - f(x)$. Find the minimum value of h on the interval $0 \leq x \leq 26$.

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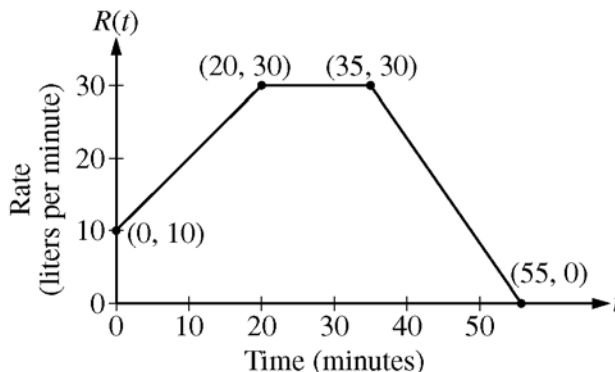
**Answer Key for AP Calculus AB
Practice Exam, Section I**

Question 1: B	Question 24: D
Question 2: A	Question 25: B
Question 3: E	Question 26: C
Question 4: A	Question 27: B
Question 5: B	Question 28: A
Question 6: B	Question 76: B
Question 7: D	Question 77: B
Question 8: C	Question 78: D
Question 9: B	Question 79: A
Question 10: A	Question 80: E
Question 11: C	Question 81: B
Question 12: A	Question 82: C
Question 13: E	Question 83: B
Question 14: D	Question 84: D
Question 15: C	Question 85: E
Question 16: C	Question 86: A
Question 17: C	Question 87: E
Question 18: B	Question 88: A
Question 19: C	Question 89: A
Question 20: C	Question 90: D
Question 21: B	Question 91: D
Question 22: A	Question 92: B
Question 23: B	

AP[®] CALCULUS AB
2015 SCORING GUIDELINES

Question 1

At time $t = 0$ minutes, a tank contains 100 liters of water. The piecewise-linear graph above shows the rate $R(t)$, in liters per minute, at which water is pumped into the tank during a 55-minute period.



- (a) Find $R'(45)$. Using appropriate units, explain the meaning of your answer in the context of this problem.
- (b) How many liters of water have been pumped into the tank from time $t = 0$ to time $t = 55$ minutes? Show the work that leads to your answer.
- (c) At time $t = 10$ minutes, water begins draining from the tank at a rate modeled by the function D , where $D(t) = 10e^{(\sin t)/10}$ liters per minute. Water continues to drain at this rate until time $t = 55$ minutes. How many liters of water are in the tank at time $t = 55$ minutes?
- (d) Using the functions R and D , determine whether the amount of water in the tank is increasing or decreasing at time $t = 45$ minutes. Justify your answer.

(a) $R'(45) = \frac{30 - 0}{35 - 55} = -\frac{3}{2}$

The rate at which water is being pumped into the tank is decreasing at $\frac{3}{2}$ liters/min² at $t = 45$ minutes.

(b)
$$\int_0^{55} R(t) dt = 20 \cdot \frac{10 + 30}{2} + 15 \cdot 30 + \frac{1}{2} \cdot 20 \cdot 30$$

$$= 400 + 450 + 300 = 1150$$

(c)
$$\text{Amt} = 100 + 1150 - \int_{10}^{55} 10e^{(\sin t)/10} dt$$

$$= 1250 - 450.275371 = 799.725 \text{ (or } 799.724)$$

(d) $R(45) = 15$
 $D(45) = 10.88815$

At time $t = 45$ minutes, the rate of water pumped into the tank is greater than the rate of water draining from the tank. Therefore, the amount of water in the tank is increasing at time $t = 45$ minutes.

2 : $\begin{cases} 1 : R'(45) \\ 1 : \text{explanation} \end{cases}$

2 : $\begin{cases} 1 : \text{sum of areas} \\ 1 : \text{answer} \end{cases}$

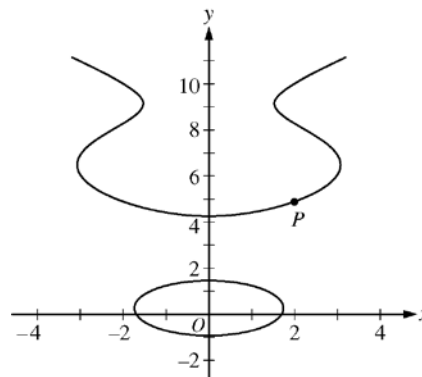
3 : $\begin{cases} 1 : \text{integral} \\ 1 : \text{expression for water in the tank} \\ 1 : \text{answer} \end{cases}$

2 : answer with justification

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2015 SCORING GUIDELINES

Question 2

The graph of the equation $x^2 = -2 + y + 5 \cos y$ is shown above for $y \leq 11$. It is known that $\frac{dy}{dx} = \frac{2x}{1 - 5 \sin y}$. The x -coordinate of point P shown on the graph is 2.



- Write an equation for the line tangent to the graph at point P .
- For $y \leq 11$, find the y -coordinate of each point on the graph where the line tangent to the graph at that point is vertical.
- Find the average value of the x -coordinates of the points on the graph in the first quadrant between $y = 5$ and $y = 9$.

(a) $x = 2 \Rightarrow y + 5 \cos y = 6 \Rightarrow y = 4.928387$

$$\left. \frac{dy}{dx} \right|_{(x,y)=(2, 4.928387)} = 0.679831$$

An equation for the line tangent to the graph at point P is
 $y = 0.680(x - 2) + 4.928$ (or $y = 0.679(x - 2) + 4.928$).

3 : $\begin{cases} 1 : y\text{-coordinate} \\ 1 : \text{slope} \\ 1 : \text{tangent line equation} \end{cases}$

(b) $1 - 5 \sin y = 0$
 $y = 0.201, 6.485$ (or 6.484), 9.223

3 : $\begin{cases} 1 : \text{sets } 1 - 5 \sin y = 0 \\ 2 : y\text{-coordinates} \end{cases}$

(c) $x = \sqrt{y + 5 \cos y - 2}$

$$\text{Average value} = \frac{1}{4} \int_5^9 \sqrt{y + 5 \cos y - 2} \, dy = 2.550$$

3 : $\begin{cases} 1 : \text{integrand} \\ 1 : \text{limits and constant} \\ 1 : \text{answer} \end{cases}$

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Question 3

t (seconds)	0	3	5	8	12
$k(t)$ (feet per second)	0	5	10	20	24

Kathleen skates on a straight track. She starts from rest at the starting line at time $t = 0$. For $0 < t \leq 12$ seconds, Kathleen's velocity k , measured in feet per second, is differentiable and increasing. Values of $k(t)$ at various times t are given in the table above.

- (a) Use the data in the table to estimate Kathleen's acceleration at time $t = 4$ seconds. Show the computations that lead to your answer. Indicate units of measure.
- (b) Use a right Riemann sum with the four subintervals indicated by the data in the table to approximate $\int_0^{12} k(t) dt$. Indicate units of measure. Is this approximation an overestimate or an underestimate for the value of $\int_0^{12} k(t) dt$? Explain your reasoning.
- (c) Nathan skates on the same track, starting 5 feet ahead of Kathleen at time $t = 0$. Nathan's velocity, in feet per second, is given by $n(t) = \frac{150}{t+3} - 50e^{-t}$. Write, but do not evaluate, an expression involving an integral that gives Nathan's distance from the starting line at time $t = 12$ seconds.
- (d) Write an expression for Nathan's acceleration in terms of t .

(a) $a(4) \approx \frac{10 - 5}{5 - 3} = \frac{5}{2} \text{ ft/sec}^2$

2 : $\begin{cases} 1 : \text{estimate} \\ 1 : \text{units} \end{cases}$

(b) $\int_0^{12} k(t) dt \approx (5)(3) + (10)(2) + (20)(3) + (24)(4) = 191 \text{ feet}$

This approximation is an overestimate since a right Riemann sum is used and the function k is increasing.

3 : $\begin{cases} 1 : \text{right Riemann sum} \\ 1 : \text{approximation with units} \\ 1 : \text{overestimate with reason} \end{cases}$

(c) $s(12) = 5 + \int_0^{12} n(t) dt$

2 : $\begin{cases} 1 : \text{integral} \\ 1 : \text{answer} \end{cases}$

(d) $n'(t) = (150)(-1)(t+3)^{-2} - 50e^{-t}(-1)$
 $= -\frac{150}{(t+3)^2} + 50e^{-t}$

2 : $n'(t)$

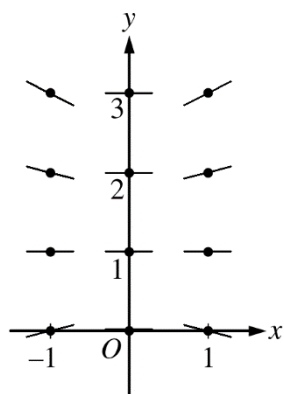
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Question 4

Consider the differential equation $\frac{dy}{dx} = \frac{x(y-1)}{4}$.

- (a) On the axes provided, sketch a slope field for the given differential equation at the twelve points indicated.
- (b) Let $y = f(x)$ be the particular solution to the differential equation with the initial condition $f(1) = 3$.
 Write an equation for the line tangent to the graph of f at the point $(1, 3)$ and use it to approximate $f(1.4)$.
- (c) Find the particular solution $y = f(x)$ to the given differential equation with the initial condition $f(1) = 3$.

(a)



2 : $\begin{cases} 1 : \text{zero slopes} \\ 1 : \text{other slopes} \end{cases}$

(b) $\left. \frac{dy}{dx} \right|_{(x,y)=(1,3)} = \frac{(1)(2)}{4} = \frac{1}{2}$

An equation for the line tangent to the graph of f at the point $(1, 3)$ is $y = \frac{1}{2}(x - 1) + 3$.

$$f(1.4) \approx \frac{1}{2}(0.4) + 3 = 3.2$$

2 : $\begin{cases} 1 : \text{tangent line} \\ 1 : \text{approximation} \end{cases}$

(c) $\frac{dy}{dx} = \frac{x(y-1)}{4}$

$$\int \frac{dy}{y-1} = \int \frac{x}{4} dx$$

$$\ln|y-1| = \frac{x^2}{8} + C$$

$$\ln 2 = \frac{1}{8} + C \Rightarrow C = \ln 2 - \frac{1}{8}$$

Because $f(1) = 3$, $y > 1$, so $|y - 1| = y - 1$.

$$\ln(y-1) = \frac{x^2}{8} + \ln 2 - \frac{1}{8}$$

$$y = 2e^{(x^2-1)/8} + 1 \text{ for all } x$$

5 : $\begin{cases} 1 : \text{separation of variables} \\ 2 : \text{antiderivatives} \\ 1 : \text{constant of integration} \\ \text{and uses initial condition} \\ 1 : \text{solves for } y \end{cases}$

Note: max 3/5 [1-2-0-0] if no constant of integration

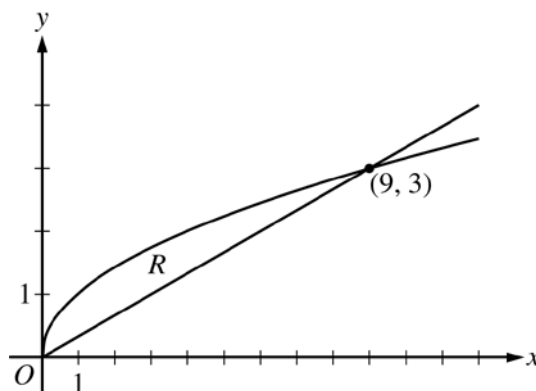
Note: 0/5 if no separation of variables

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Question 5

Let R be the region in the first quadrant enclosed by the graphs of $g(x) = \sqrt{x}$ and $h(x) = \frac{x}{3}$, as shown in the figure above.

- Find the area of region R .
- Write, but do not evaluate, an expression involving one or more integrals that gives the volume of the solid generated when R is revolved about the horizontal line $y = 4$.
- Find the maximum vertical distance between the graph of g and the graph of h between $x = 0$ and $x = 16$. Justify your answer.



$$\begin{aligned} \text{(a) Area} &= \int_0^9 \left(\sqrt{x} - \frac{x}{3} \right) dx = \left[\frac{2}{3} x^{3/2} - \frac{1}{6} x^2 \right]_0^9 \\ &= \frac{2}{3} \cdot 27 - \frac{1}{6} \cdot 81 = \frac{9}{2} \end{aligned}$$

3 : $\begin{cases} 1 : \text{integrand} \\ 1 : \text{antiderivative} \\ 1 : \text{answer} \end{cases}$

$$\text{(b) Volume} = \pi \int_0^9 \left[\left(4 - \frac{x}{3} \right)^2 - \left(4 - \sqrt{x} \right)^2 \right] dx$$

3 : $\begin{cases} 2 : \text{integrand} \\ 1 : \text{limits and constant} \end{cases}$

(c) Consider the function $D(x) = \sqrt{x} - \frac{x}{3}$.

$$D'(x) = \frac{1}{2} x^{-1/2} - \frac{1}{3} = \frac{1}{2\sqrt{x}} - \frac{1}{3}$$

$$D'(x) = 0 \Rightarrow x = \frac{9}{4}$$

3 : $\begin{cases} 1 : \text{sets } D'(x) = 0 \\ 1 : \text{identifies } x = \frac{9}{4} \text{ as a candidate} \\ 1 : \text{answer and justification} \end{cases}$

x	$D(x)$	Distance between graphs
0	0	0
$\frac{9}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
16	$-\frac{4}{3}$	$\frac{4}{3}$

The maximum vertical distance between the graph of g and the graph of h between $x = 0$ and $x = 16$ is $\frac{4}{3}$.

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Question 6

Let $g(x) = 4(x+1)^{-2/3}$ and let f be the function defined by $f(x) = \int_0^x g(t) dt$ for $x \geq 0$.

- (a) Find $f(26)$.
- (b) Determine the concavity of the graph of $y = f(x)$ for $x > 0$. Justify your answer.
- (c) Let h be the function defined by $h(x) = x - f(x)$. Find the minimum value of h on the interval $0 \leq x \leq 26$.

(a)
$$f(26) = \int_0^{26} 4(x+1)^{-2/3} dx = \left[4 \cdot 3(x+1)^{1/3} \right]_{x=0}^{x=26}$$

$$= 36 - 12 = 24$$

3 : $\begin{cases} 2 : \text{antiderivative} \\ 1 : \text{answer} \end{cases}$

(b)
$$f'(x) = g(x) = 4(x+1)^{-2/3}$$

$$f''(x) = 4 \left(-\frac{2}{3} \right) (x+1)^{-5/3} = -\frac{8}{3(x+1)^{5/3}}$$

$$f''(x) < 0 \text{ for } x > 0.$$

3 : $\begin{cases} 1 : f'(x) \\ 1 : f''(x) \\ 1 : \text{answer with justification} \end{cases}$

The graph of $y = f(x)$ is concave down for $x > 0$.

(c)
$$h'(x) = 1 - f'(x) = 1 - 4(x+1)^{-2/3}$$

$$h'(x) = 0 \Rightarrow 1 = \frac{4}{(x+1)^{2/3}} \Rightarrow x = 7$$

3 : $\begin{cases} 1 : \text{sets } h'(x) = 0 \\ 1 : \text{identifies } x = 7 \text{ as a candidate} \\ 1 : \text{answer with reason} \end{cases}$

$$h(7) = 7 - \int_0^7 4(x+1)^{-2/3} dx = 7 - \left[4 \cdot 3(x+1)^{1/3} \right]_{x=0}^{x=7}$$

$$= 7 - [24 - 12] = -5$$

$$h(26) = 26 - 24 = 2$$

x	$h(x)$
0	0
7	-5
26	2

The minimum value of h on the interval $0 \leq x \leq 26$ is -5 .